

5.11 Solving Problems Involving Ratios

Lesson Introduction

 Benchmark	MA.6.AR.3.5 Solving mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.		 Learning Target	I can solve problems involving ratios, rates, and unit rates.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.M.1.1 Solve multi-step real-world problems that involve converting measurement units to equivalent measurements within a single system of measurement.		MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.	
 Essential Questions	- How do you solve problems involving ratios, rates, and unit rates? - How do you convert units of measurement? - How do you calculate and compare rates? - Why is it important to find the unit rate when comparing different ratios?			
 Lesson Narrative	In this lesson, students work collaboratively to solve problems involving ratios, rates, and unit rates at three stations. At station 1, students begin by exploring whether various offers are a good deal based on the original and new price. Station 2 allows students to continue applying skills from the unit to convert measurements in a recipe from tablespoons to cups and vice-versa. At station 3, two texters are trying to determine who is the quickest; students must find and apply unit rates of each to declare a winner. This lesson culminates in a self-assessment to explain what students know and what they still have questions about.			
 Component Summary	5.11.1 Warm-Up (~5 min.)	Students determine if they can complete a task in a certain amount of time if they work at a given rate. (SE p. 225)	5.11.3 Activity (10-12 min.)	Students work collaboratively in station activity to solve problems involving ratios, rates, and unit rates. (SE p. 227)
	5.11.2 Activity (10-12 min.)	Students work collaboratively in station activity to solve problems involving ratios, rates, and unit rates. (SE p. 225)	5.11.4 Activity (10-12 min.)	Students work collaboratively in station activity to solve problems involving ratios, rates, and unit rates. (SE p. 227)
	5.11.5 Wrap-Up (~5 min.)	Students engage in meaningful self-reflection on growth mindset and learning progression from the lesson and the unit. (SE p. 229)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Scientific Calculators - Cards for Station 1 - Devices for Station 3	
			Additional Preparation	
			Station 1: Print and cut for blackline master activity; remove Task Card F	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may divide how many items there are by the price instead of dividing the price by how many items there are when finding the unit price.	The lesson has an Activity that consists of three different stations. Be sure to keep track of the time and ask students to rotate through the stations. Also, consider how students will check their answers. You could debrief all three stations at the end or have a key for students to self-check at each station.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display Students can engage in a Collect and Display for their stations in the Activity. Using chart paper to represent their strategies, approaches, and visual models provides students the opportunity to synthesize their learnings from the unit as they apply what they know in practice. Students can then engage in a Gallery Walk after to engage with the thinking of others.	MA.K12.MTR.7.1: Real-World Contexts Students solve problems connected to real-world contexts in the Activity. Students see how calculating ratios, rates, and unit rates can help them determine the best value when shopping, convert ingredient measurements when cooking or baking, and compare rates. Each real-world context should spark student interest and motivate them to engage in solving the problems.
 Differentiate Instruction	The Activity in the lesson consists of three stations at which students are applying concepts of ratios in a variety of real-world contexts. Consider a variety of strategies for engaging your students with Station Rotations. - Each small group completes each of the three stations. For small groups of 3-4 students per group, consider having 3 copies of each station for a total of 9 groups that could work simultaneously. After all groups have completed all three stations, have one student from each group pair up with a different group and share responses to verify accuracy and compare strategies. - Each small group is assigned one station at which they are experts, creating an anchor chart to display their work. After each group completes their work, every student then pairs with a student from another group and both students then act as experts or teachers of their station. Students repeat this process for the third station.	
Lesson Notes:		

5.11.1 Warm-Up: Bell Work

Component Overview: Students determine if someone will be able to complete a task in a certain amount of time if they work at a given rate.



5.11.1 Warm-Up: Bell Work

1. Mai has to fill and seal 60 packages of homemade trail mix to sell at the upcoming farmer's market. She filled and sealed 25 packages in 32 minutes. Mai wonders, "If I keep working at this rate, will I be able to get all of these done in less than 1 hour?"

Explain whether Mai will be able to get all 60 packages filled in less than 1 hour working at this rate.

The first step in determining if Mai would be able to fill and seal 60 packages in less than an hour, is to determine how much time it takes for one package to be filled and sealed.

$$\frac{25 \text{ packages}}{32 \text{ minute}} = 0.78125 \text{ packages per minute}$$

The next step is to multiply the packages per minute by 60 to determine the amount of time it will take to fill and seal 60 packages.

$$0.78125 \cdot 60 = 46.875 \approx 47 \text{ packages}$$

If Mai keeps working at this rate, she will not finish all 60 packages in less than an hour.



Consider allowing students to discuss their thinking before writing their answers to engage auditory learners. Encourage students to use a table or double number line to help them solve. This provides an entry point for visual learners.

5.11.2 Activity: Station 1: Is It a Deal?

Component Overview: Students work collaboratively in the three following station activities to solve problems involving ratios, rates, and unit rates. The class can be broken into six small groups with two versions of each station. Students can either rotate through the stations or the stations can be passed from group-to-group.



5.11.2 Activity: Station 1: Is It a Deal?

Your teacher will give you a set of cards showing different offers.

- Find Card A and work with your partner to decide whether the offer on Card A is a good deal. Explain or show your reasoning in the space provided.

Card A

$$\text{Original Cost per Water} = \frac{3.16}{4} = \$0.79$$

$$\text{New Deal Cost per Water} = \frac{2.25}{3} = \$0.75$$

The original cost of the 4 pack of water is 4 cents more, so the new deal a better deal.

- Next, split cards B–E so you and your partner each have two cards.
 - Decide individually if your two cards are good deals. Explain your reasoning for your two cards in the table.

Is the new deal a good deal?			
Card B	Card C	Card D	Card E
<i>Original</i> $= \frac{3.50}{10} = \$0.35$ <i>New Deal</i> $= \frac{2.40}{6} = \$0.40$ <i>The new deal is not the better deal because the cost is lower in the original price.</i>	<i>Original</i> $= \frac{4.40}{5} = \$0.88$ <i>New Deal</i> $= \frac{3.12}{4} = \$0.78$ <i>The new deal is the better deal because the cost is lower than the original price.</i>	<i>Original</i> $= \frac{14.40}{16} = \$0.90$ <i>New Deal</i> $= \frac{9.00}{10} = \$0.90$ <i>Both deals have the same unit price, so neither is a better deal than the other.</i>	<i>Original</i> $= \frac{6.80}{8} = \$0.85$ <i>New Deal</i> $= \frac{5.22}{6} = \$0.87$ <i>The new deal is not the better deal because the original cost is lower than the new deal.</i>



All three stations provide an opportunity for multiple entry points and engagement of different learning styles.

Each station is set up to involve students in mathematical discussion with their partners or small groups, providing an entry point for auditory learners.

Each station also allows students to choose their own methods of representation. The visuals provided for Stations 1 and 3 offer an entry point for visual learners as well. Consider how to best engage kinesthetic learners as well, through the use of manipulatives or another hands-on strategy, like texting at Station 3.



Consider how to best facilitate student discussion at each station to ensure students are engaging with the material and deepening their understanding.

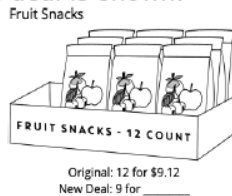


Circulate as students are working in their pairs or small groups and listen for discourse centered on using unit rates to determine the best deals, fastest texter, etc. If you notice students are not making this connection, ask probing questions to guide student thinking like:

- How can you use what you know about ratios and unit rates to find the best deal?
- How will you know which deal is the best? Should the unit rate be larger or smaller than the original price?
- If you find the unit rate, how can you use it to find the cost of the item?

5.11.2 Activity: Station 1: Is It a Deal? (continued)

- b. For each of your cards, explain to your partner if you think it is a good deal and why.
The answers for cards B-E are in the above table.
 - c. Listen to your partner's explanations for their cards. If you disagree, explain your thinking. Write your partner's explanation for their cards in the table above, after resolving any differences.
Answers will vary. Sample responses in the above table.
 - d. Revise any decisions about your cards based on the feedback from your partner.
3. When you and your partner are in agreement about cards B-E, place all the cards you think are a good deal in one stack and all the cards you think are a bad deal in another stack.
 4. Another potential deal is shown.



- a. What is the price for 9 packs of fruit snacks at the same unit price as the original?

$$\text{Original} = \frac{9.12}{12} = \$0.76$$

$$0.76 \cdot 9 = \$6.84$$

- b. Complete the statement.

Any price less than \$ 6.50 for 9 packs of fruit snacks would be considered a good deal because the cost per pack would be less than \$ 0.72.

Answers will vary. Any price less than or equal to \$6.84 will be a better deal.



If students have not mastered multiplication and division facts, let them use a calculator to solve the ratio, rate, and unit rate problems. The calculator will allow students to focus on mastering the learning goals of the lesson. Require students to write each calculation they use the calculator to find. This will reinforce their skills.

5.11.3 Activity: Station 2: Cooking with a Tablespoon

Component Overview: Students work collaboratively in this station activity (#2 of 3).



5.11.3 Activity: Station 2: Cooking with a Tablespoon

Diego is trying to follow a recipe, but he cannot find any measuring cups! He only has a tablespoon.

In the cookbook, it says that 1 cup equals 16 tablespoons.

- How could Diego use the tablespoon to measure out these ingredients?

- $\frac{1}{2}$ cup almonds

Note: 1 tablespoon (Tbsp)

$$\frac{1}{2}c \times \frac{16 \text{ Tbsp}}{1c} = 8 \text{ Tbsp of almonds}$$

- $1\frac{1}{4}$ cup of oatmeal

Note: $1\frac{1}{4} = \frac{5}{4}$

$$\frac{5}{4}c \times \frac{16 \text{ Tbsp}}{1c} = 20 \text{ Tbsp of oatmeal}$$

- $2\frac{3}{4}$ cup of flour

Note: $2\frac{3}{4} = \frac{11}{4}$

$$\frac{11}{4}c \times \frac{16 \text{ Tbsp}}{1c} = 44 \text{ Tbsp of flour}$$

- Diego also adds the following ingredients. How many cups of each did he use?

- 28 tablespoons of sugar

$$\frac{1c}{16 \text{ Tbsp}} \times \frac{28 \text{ Tbsp}}{1} = \frac{7}{4} = 1\frac{3}{4}c \text{ of sugar}$$

- 6 tablespoons of cocoa powder

$$\frac{1c}{16 \text{ Tbsp}} \times \frac{6 \text{ Tbsp}}{1} = \frac{3}{8}c \text{ of cocoa powder}$$



Think-Pair-Share

Consider structuring station 2 with the Think-Pair-Share routine. Consider including instructions or guidance for this routine either at the station itself if students are moving around the room, or on a white board that students can reference as they are working through each station.



This section requires several conversions using different ratios involving fractions. Some students may confuse how to write the ratio correctly, perhaps inverting the ratio, or they may write the mixed numbers as improper fractions incorrectly.

To address misconceptions, ask clarifying questions:

- If you are converting cups to tablespoons, will the equivalent amount result in more or less tablespoons? How do you know?
- What ratio represents this? How can you write this ratio using a fraction bar?
- If you are converting tablespoons to cups, will the equivalent amount result in more or fewer cups? How do you know?
- What ratio represents tablespoons to cups? How can you write this ratio using a fraction bar?

5.11.4 Activity: Station 3: Turbo Texting

Component Overview: Students work collaboratively in this station activity (#3 of 3).

5.11.4 Activity: Station 3: Turbo Texting

1. Jonah was with his friend Erin after school and he needed to text his mom. Erin watched Jonah text and commented on how slow he was.

They decided to have a text race to determine who was faster. The image below shows what each person texted in the race.

I am the fastest texter!!! You are so slow! You will never beat me!



Think so? You're about to lose this! It's on!!

- a. Discuss any initial observations with your partner.

Answers will vary. Jonah's is longer than Erin's.

- b. Erin was able to send her text slightly faster than Jonah. Explain if you think this means she is the faster texter.

Answers will vary. Erin's text did not have as many characters as Jonah's text, so she was able to send her text quicker. This does not make Erin the faster texter because she had fewer characters in her text.

- c. Describe the relationship you notice between the number of characters in a text versus how long it might take to type it.

Answers will vary. The longer the text message, the longer it takes to send. Based on the number of characters listed, Jonah's text may take a little longer.



Advanced learners can be challenged to determine their texting rate and compare it to Jonah's, Erin's, or their classmates.



This component is centered on discussion and inquiry.

The discourse for question 1 should therefore sound more informal and inquisitive as students make predictions and assumptions before having concrete numbers to work with, leading them to determine what information they will need moving forward to conclude who is the faster texter.

5.11.4 Activity: Station 3: Turbo Texting (continued)

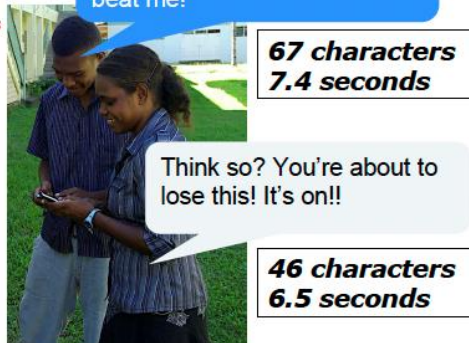
- d. What additional information would be helpful to determine who is the faster texter?

Answers will vary. To determine who is the faster texter, we would need to know how long it took both Jonah and Erin to send their texts.

2. The image below gives additional information about each person's text.

*Jonah: $\frac{67}{7.4} \approx 9.05 \approx$
9 characters per
second*

*Erin: $\frac{46}{6.5} \approx 7.08 \approx$
7 characters per
second*



Jonah is the faster texter because he sends approximately 2 more characters per second.

Work with your partner or small group to determine who is the faster texter, using your preferred strategy. Show your work.

Work shown above.



Some students may not realize they also need to know the amount of time it took each person to text to truly know who the faster texter is, assuming that whoever texted quickest must be the fastest. Ask guiding questions to ensure students arrive at this conclusion before completing the rest of the section:

- *Whose text is longer? How do you know?*
- *If Jonah and Erin send different length texts in different amounts of time, how can you determine who is the fastest?*
- *How can you apply what you know about unit rates to solve this problem?*

5.11.4 Activity: Station 3: Turbo Texting (continued)

3. Jonah and Erin decided to send one last text to see who could send it out faster. The image shows the text they each sent.

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

154 characters



Encourage students to use any representation they have learned throughout this unit to solve. Student choice allows for multiple entry points and can also encourage students to make connections across representations as they work with their partners or groups to solve.

If both people sent the message above at the same rate as before, predict about how many seconds it took each person to text the message to the nearest tenth of a second. Show your work.

Brother	Predicted Time
Jonah	17.0 sec.
Erin	21.8 sec.

Jonah:

$$\frac{154 \text{ characters}}{1} \times \frac{7.4 \text{ sec}}{67 \text{ characters}} = \frac{1,139.6}{67} \approx 17.01 \text{ sec.}$$

Erin:

$$\frac{154 \text{ characters}}{1} \times \frac{6.5 \text{ sec}}{46 \text{ characters}} = \frac{1,001}{46} \approx 21.76 \text{ sec.}$$

5.11.5 Wrap-Up: Reflection

Component Overview: Students reflect on a question they have, a mistake they made, something they are unsure about, and something they are confident in from the lesson. Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson.



5.11.5 Wrap-Up: Reflection

- Complete the learning period based on today's lesson.

Answers will vary.

One question I would like to have answered ...	
One mistake I learned from today ...	One thing I am not sure about ... YET!
One thing I am confident I know is ...	



What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of ratio and rate concepts addressed throughout the unit and how to best to address potential gaps in understanding prior to the unit assessment. Consider making notes on the unit guide for things you would want to keep in mind for this unit next school year.